CLINICAL TRIAL FAILURES · DISCONTINUATION RATE · ONCOLOGY PHASE 2,3 · STARTS 2015–2024

HER3: 4 molecules stopped early

5 stopped trials across 4 molecules: Tarloxotinib, Cdx-3379, Patritumab and Patritumab Deruxtecan. Against 5.2% for all oncology trials with a resolved drug target and 4.5% across all oncology Phase 2,3 trials in the same window. The rate counts trials that stopped early for efficacy, safety or benefit–risk reasons. Trials that ran to completion and missed their primary endpoint are counted separately below, never in the rate.

16.7%
5 of 30 closed trials (95% CI 7.3%–33.6%)

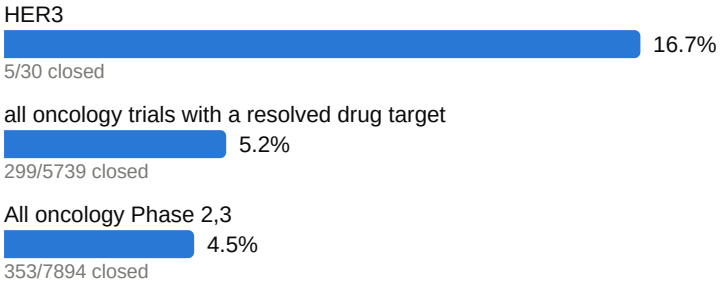
4 / 0 / 0
efficacy / safety / both (adds to 5)

33.0%
of 91 trials have closed · 61 still open or unresolved

7.8%
stopped for a biological reason within 3 years of starting (95% CI 3.2%–18.2%; 30 still at risk; 1.5% area-wide)

— **Completed and missed the primary endpoint.** None of the 12 completed trials in this segment posted a primary result on ClinicalTrials.gov that can be read, so nothing can be said here either way. 8 of the 12 completed trials that finished more than a year ago have posted no results at all — a gap in what can be known, not a sign of failure.

How this compares



What is in this class: 96 distinct experimental drugs across 91 trials (small molecule 48, adc 47, antibody 40, unknown 16). Most tested: IZALONTAMAB BRENGITECAN (23), PATRITUMAB DERUXTECAN (15), POZIOTINIB (10), IZALONTAMAB (10), DANVILOSTOMIG (10). A class groups drugs by what they act on; check that the grouping is one you would make before reading the rate as a property of the mechanism.

How much does this rest on one decision? The 5 stops came from 4 sponsor-asset programmes across 3 sponsors; the largest (Rain Oncology Inc) contributed 2; removing that programme's trials from both sides leaves 11.1%. Screened alongside 85 other segments, the rate is within what screening that many classes could produce (q=1); the molecules and trials that failed here are the record either way. A further 2 closed trials here were terminated with no cause recorded in the registry; if every one of them were biological the rate would be 23.3%. Read it as a screen worth checking against the underlying trials.

Bars show the share of closed trials stopped for biological reasons. Confidence intervals overlap where sample sizes are small — read the counts, not just the bars.

By start cohort

Trials started	Stops	Closed	Rate	95% CI
2015–2017	3	15	20.0%	7.0%–45.2%
2018–2020	2	8	25.0%	7.1%–59.1%
2021–2024	0	7	0.0%	0.0%–35.4%

Where the stops sit

Efficacy / futility: 4, Benefit–risk: 1.
Sponsors with most stops: Daiichi Sankyo (2), Rain Oncology Inc (2), Celldex Therapeutics (1).

The stopped trials

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT04479436	2	Daiichi Sankyo	PATRITUMAB DERUXTECAN	2022-02	Efficacy	Study was terminated early given the Interim Analysis for Part 1 (signal finding) did not meet pre-specified criteria and will not proceed to Part 2. Sponsor will proceed closing the study.
NCT03254927	2	Celldex Therapeutics CLDX	CDX-3379, CETUXIMAB	2020-09	Benefit–risk	Sponsor decision based on emerging risk-benefit profile

Trial	Ph	Sponsor	Experimental drugs	Stopped	Type	Registry stop reason
NCT02633800	2	Daiichi Sankyo	PATRITUMAB	2018-01	Efficacy	Trial was terminated by sponsor due to lack of efficacy.
NCT02454842	2	Rain Oncology Inc	TARLOXOTINIB	2017-01	Efficacy	Lack of efficacy
NCT02449681	2	Rain Oncology Inc	TARLOXOTINIB	2017-01	Efficacy	Lack of efficacy

Method

Denominator: ClinicalTrials.gov interventional Phase 2,3 oncology trials started 2015–2024 that have closed (completed or terminated). Numerator: terminated trials whose registry stop reason is classified as biological (efficacy, safety or benefit–risk) by Classification V2 — held-out precision 95.5%, recall 95.3% (n=600). Drugs are linked to ChEMBL and the NCI Thesaurus; 76% of industry oncology trials in this window carry a resolved drug target, and a trial without one cannot enter a mechanism class. Intervals are Wilson 95%.

Limits

Not a failure rate: completed trials that missed their endpoints are not in the rate. They are counted separately where the sponsor posted a primary result that can be read — most did not, so that count is a floor — and programmes discontinued after a completed trial do not appear. Stop reasons are sponsor-reported. This is a closed-trial proportion, not a time-to-event analysis: only 33.0% of trials in this segment have closed, and a trial that stops early enters the denominator sooner than one that runs to completion, which can inflate the rate in immature segments. Recent cohorts have fewer closed trials, so their rates are less stable. Research signals, not clinical or investment advice.

The whole cohort, trial by trial. The diligence report names every trial behind these figures — the ones still running and the ones that completed and missed their primary endpoint included — and places your molecule against each one that failed. €99, delivered immediately: **clinicaltrialfailures.com/asset-check**

Sources: ClinicalTrials.gov (NLM); ChEMBL (EMBL-EBI, CC BY-SA 3.0); NCI Thesaurus (NCI); RxNorm/RxClass (NLM); SEC EDGAR. Classification, linkage and rates by Clinical Trial Failures. Rebuilt weekly; this brief covers trials started 2015–2024.